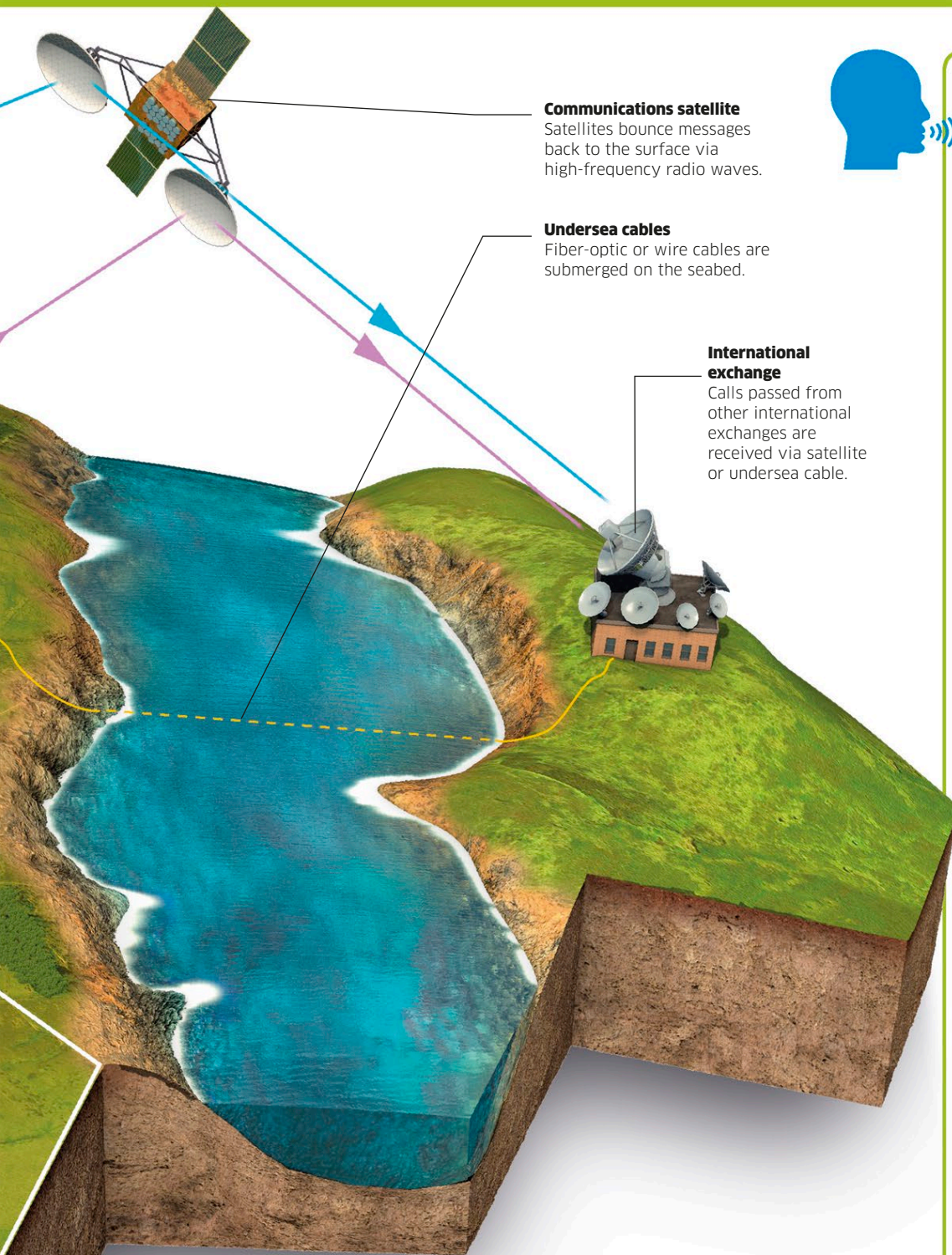
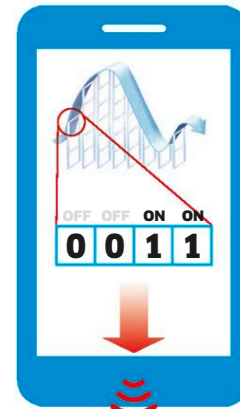




Communications satellite
Satellites bounce messages back to the surface via high-frequency radio waves.

Undersea cables
Fiber-optic or wire cables are submerged on the seabed.

International exchange
Calls passed from other international exchanges are received via satellite or undersea cable.

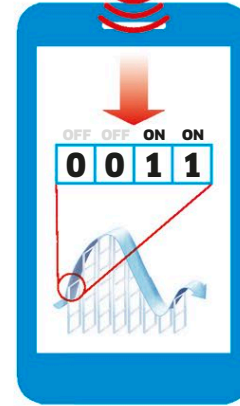


Speech conversion
Our voices are converted from analog signals to digital ones to make calls.

1 A cell phone captures sound as a continuously varying or analog signal. The signal is measured at various points and each point is given a value. Here a point on the signal is measured as 3, which is shown as its binary equivalent of 0011. The phone's analog to digital converter produces strings of these binary numbers (see p.95).



2 The 1s and 0s of the binary number 0011 become off/off/on/on. The phone transmits the on/off values, encoding them as sudden changes to the signal's waves. The signal passes from base station to mobile exchange to base station.



3 The phone receives the digital signal and interprets the on/off transmission as strings of binary numbers. The phone's digital to analog converter turns the binary numbers back into analog information.



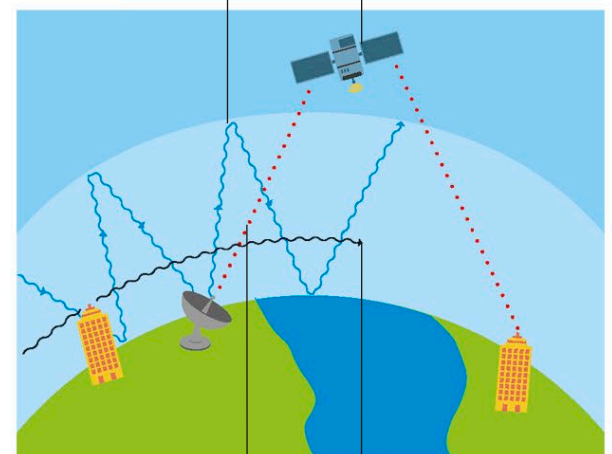
4 The phone's speaker sends an analog signal we hear as a sound wave.

The ionosphere and radio waves

The ionosphere is a region of the atmosphere that contains ions and free electrons. This causes it to reflect some lower-frequency, longer-wavelength radio waves over large distances.

Short and medium waves are reflected off the top of the ionosphere and the Earth's surface.

Communications satellites retransmit signals to distant areas.



High-frequency signals pass through the ionosphere to reach satellites.

Long waves can travel as a direct surface wave for thousands of miles.

Telecommunications

Modern telecommunications use electricity, light, and radio as signal carriers. The global telephone network enables us to communicate worldwide, using radio links, fiber-optic cables, and metal cables.

Signals representing sounds, images, and other data are sent as either analog signals, which are unbroken waves, or as digital signals that send binary code as abrupt changes in the waves. Radio waves transmit radio and TV signals through the air around Earth, while microwave wavelengths are used in cell phones, Wi-Fi, and Bluetooth. Cables carry signals both above and below ground—as electric currents along metal wires, or as pulses of light that reflect off the glass interiors of fiber-optic cables.